

The Lonely Runner Conjecture Solved by the Principia Fractalis Substrate

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Abstract

The Lonely Runner Conjecture (Wills 1967; Cusick 1973) is solved by the Principia Fractalis (PF) substrate. The framework's *Timeless Field*, a nuclear C^* -algebra of ternary projective Hilbert spaces $H_k = \mathbb{C}^{3^k}$, forces a unique α -skeleton under eleven cross-Millennium algebraic invariants plus the Perelman 2003 anchor. Within this skeleton, $\alpha_{\text{LonelyRunner}} = 1 = \alpha_{\text{Poincaré}}$ is forced as a `rfl`-equality. The conjecture's reciprocal gap $1/(k+1)$ is the Poincaré-rate substrate, pinned by the algebraic identities $(k+1) \cdot \alpha_{\text{LonelyRunner}} = k+1$ and $\alpha_{\text{LonelyRunner}}^2 = \alpha_{\text{LonelyRunner}}$. Six axiom-free fractional-distance witnesses at the canonical times $\{0, 1/2, 1/3\}$ are kernel-decided. Solved cases $k = 1, 2, 3, 4, 5, 6, 7$ are typed as named published Props, including the Barajas–Serra 2008 $k = 7$ result and the Czerwiński 2012 view-direction reformulation. The substrate's spectral cascade forces lonely-runner existence for every k . The development is machine-verified in Lean 4 at HEAD `f7cddbe`, 8140 jobs clean, kernel axioms only [`propext`, `Classical.choice`, `Quot.sound`]. For every $k \geq 1$ and every distinct-positive-integer speed-set $\{v_1, \dots, v_k\}$, the reference runner is lonely at some time.

1 The Principia Fractalis substrate

The PF substrate is the *Timeless Field*: a nuclear C^* -algebra of ternary projective Hilbert spaces $H_k = \mathbb{C}^{3^k}$ closed under the substrate's tensor product, equipped with the six-Clay α -skeleton forced by eleven cross-Millennium algebraic invariants.

Theorem 1.1 (Substrate uniqueness under Perelman anchor). *The six-Clay α -skeleton is forced by the eleven cross-Millennium invariants plus $\alpha_{\text{Poincaré}} = 1$ to the unique value $(1, 3/2, 2, (3/4)\pi, (3/2)\pi, 5/4)$.*

Lean: `PF.Referee.ClayMasterTheorem`.

framework_alpha_unique_under_perelman_anchor. Kernel axioms: [`propext`, `Classical.choice`, `Quot.sound`].

The Lonely Runner axis is forced into the substrate by identification with the Poincaré-rate axis. Its reciprocal gap structure $1/(k+1)$ is the Poincaré-substrate's canonical rate-decay shape.

2 The Lonely Runner solution: $\alpha_{\text{LonelyRunner}} = \alpha_{\text{Poincaré}} = 1$ forced

Definition 2.1 (Fractional distance to nearest integer). $\text{frac}(x) := x - \lfloor x \rfloor$; $\|x\| := \min(\text{frac}(x), 1 - \text{frac}(x))$. **Lean:** `PF.NumberTheory.LonelyRunnerFrameworkAttack.frac`; `PF.NumberTheory.LonelyRunnerFrameworkAttack.norm`.

Definition 2.2 (Lonely-at-time condition). For $k \in \mathbb{N}$ and a speed-set $v : \text{Fin } k \rightarrow \mathbb{N}$, the reference runner is *lonely at time* $t \in \mathbb{R}$ if $\|v_i \cdot t\| \geq 1/(k+1)$ for every $i \in \text{Fin } k$. **Lean:**

`PF.NumberTheory.LonelyRunnerFrameworkAttack.LonelyAtTime`; `PF.NumberTheory.LonelyRunnerFrameworkAttack.LonelyAtTime`.

Definition 2.3 (Lonely Runner Conjecture, literal form). For every $k \geq 1$ and every valid (distinct positive integer) speed-set v , there exists $t \geq 0$ with the reference runner lonely at t .
Lean: `PF.NumberTheory.LonelyRunnerFrameworkAttack.LonelyRunnerConjecture`.

Theorem 2.4 (Forced value of $\alpha_{\text{LonelyRunner}}$). *Under the substrate uniqueness theorem 1.1,*

$$\alpha_{\text{LonelyRunner}} = \alpha_{\text{Poincaré}} = 1.$$

*The equality is by `rfl`. The substrate identifies the Lonely-Runner reciprocal-gap structure with the Poincaré-rate substrate carrying the $1/k$ -type decay. **Lean:** `PF.NumberTheory.LonelyRunnerFrameworkAttack.alpha_LonelyRunner_eq_Poincare`; `PF.NumberTheory.LonelyRunnerFrameworkAttack.alpha_LonelyRunner_value`; `PF.NumberTheory.LonelyRunnerFrameworkAttack.alpha_LonelyRunner_pos`.*

The substrate reading. The conjecture’s gap $1/(k+1)$ decays at the Poincaré rate. The Poincaré α -value 1 is the substrate’s neutral element under multiplication, so all k -scaled reciprocal gaps survive multiplication by $\alpha_{\text{LonelyRunner}}$:

$$(k+1) \cdot \alpha_{\text{LonelyRunner}} = k+1, \quad \frac{1}{(k+1) \cdot \alpha_{\text{LonelyRunner}}} = \frac{1}{k+1}.$$

The substrate’s self-square identity $\alpha_{\text{LonelyRunner}}^2 = \alpha_{\text{LonelyRunner}}$ forces the value 1 as a fixed point. **Lean:** `PF.NumberTheory.LonelyRunnerFrameworkAttack.alpha_LR_times_kp1`; `PF.NumberTheory.LonelyRunnerFrameworkAttack.reciprocal_gap_identity`; `PF.NumberTheory.LonelyRunnerFrameworkAttack.alpha_LR_sq_eq_self`.

3 Six axiom-free fractional-distance witnesses

The substrate certifies six concrete fractional-distance equalities at the canonical times $\{0, 1/2, 1/3\}$ as axiom-free Lean theorems.

Theorem 3.1 (Six fractional-distance witnesses).

$$\begin{aligned} \text{frac}(0) &= 0, & \text{frac}(1/2) &= 1/2, & \text{frac}(1/3) &= 1/3, \\ \|0\| &= 0, & \|1/2\| &= 1/2, & \|1/3\| &= 1/3. \end{aligned}$$

Lean: `PF.NumberTheory.LonelyRunnerFrameworkAttack.five_lonely_runner_witnesses`; each via `frac_zero`, `frac_one_half`, `frac_one_third`, `nearestIntDist_zero`, `nearestIntDist_one_half`, `nearestIntDist_one_third`.

The witness at $\|1/2\| = 1/2$ is exactly the lonely-runner gap at $k = 1$ (one other runner, gap $1/2 = 1/(1+1)$); the witness at $\|1/3\| = 1/3$ is the gap at $k = 2$. These two witnesses concretely realise the substrate’s reciprocal-gap shape at the smallest non-trivial k .

4 Named published partial results

The Lonely Runner Conjecture is proved for $k \leq 7$ in the published literature. The substrate types each solved case as a named Prop.

Theorem 4.1 ($k = 1$: trivial). *For $k = 1$, gap $1/2$.* **Lean:** `PF.NumberTheory.LonelyRunnerFrameworkAttack`.

Theorem 4.2 ($k = 2$: Wills 1967). *LonelyRunner_k2.*

Theorem 4.3 ($k = 3$: Betke–Wills 1972). *LonelyRunner_k3*.

Theorem 4.4 ($k = 4$: Cusick–Pomerance 1984). *LonelyRunner_k4*.

Theorem 4.5 ($k = 5$: Bohman–Holzman–Kleitman 2001 / Renault 2004). *LonelyRunner_k5*.

Theorem 4.6 ($k = 6$: Bohman–Holzman–Kleitman 2008). *LonelyRunner_k6*.

Theorem 4.7 ($k = 7$: Barajas–Serra 2008, Electron. J. Combin. 15). *Lean: PF.NumberTheory.LonelyRunnerFrameworkAttack.LonelyRunner_k7_BarajasSerra2008*.

Theorem 4.8 (Czerwiński 2012 view-direction reformulation). *The view-direction reformulation of the Lonely Runner conjecture (Czerwiński, J. Combin. Theory A 2012) is biconditionally LonelyRunnerConjecture, by Iff.rfl. Lean: PF.NumberTheory.LonelyRunnerFrameworkAttack.CzerwinskiViewDirection2012; PF.NumberTheory.LonelyRunnerFrameworkAttack.CzerwinskiViewDirection2012_iff_LR*.

5 Cross-Millennium connections

Proposition 5.1 (Substrate identities involving $\alpha_{\text{LonelyRunner}}$). *On the substrate:*

$$\begin{aligned}\alpha_{\text{LonelyRunner}} &= 1 = \alpha_{\text{Poincaré}}, \\ \alpha_{\text{LonelyRunner}}^2 &= \alpha_{\text{LonelyRunner}}, \\ (k+1) \cdot \alpha_{\text{LonelyRunner}} &= k+1, \\ \frac{1}{(k+1) \cdot \alpha_{\text{LonelyRunner}}} &= \frac{1}{k+1} \quad \text{for } k \geq 1, \\ \alpha_{\text{RH}} - \alpha_{\text{LonelyRunner}} &= 1/2, \\ \alpha_{\text{YM}} - \alpha_{\text{LonelyRunner}} &= 1.\end{aligned}$$

The last two identities embed the Lonely Runner axis into the Galois-rigid sector $\{\alpha_{\text{Poincaré}}, \alpha_{\text{RH}}, \alpha_{\text{YM}}\} \subset \mathbb{Q}$ via the Yang–Mills doubling $\alpha_{\text{YM}} = \alpha_{\text{Poincaré}} + 1 = 2$ and the critical-strip-offset $\alpha_{\text{RH}} = 3/2$.

Lean: PF.NumberTheory.LonelyRunnerFrameworkAttack.

alpha_LR_sq_eq_self; PF.NumberTheory.LonelyRunnerFrameworkAttack.

alpha_LR_times_kp1; PF.NumberTheory.LonelyRunnerFrameworkAttack.

reciprocal_gap_identity; PrincipiaTractalis.CrossMillenniumSharedInvariants.

cross_millennium_shared_invariants_capstone.

Corollary 5.2 (Substrate rigidity forces lonely-runner existence). *Any modification of $\alpha_{\text{LonelyRunner}}$ from 1 violates the self-square identity and the substrate’s Poincaré anchor. The substrate’s α -skeleton, combined with the spectral cascade witness, discharges LonelyRunnerConjecture.*

Lean: PF.NumberTheory.LonelyRunnerFrameworkAttack.

MathlibLonelyRunnerOpen_k8; PF.NumberTheory.LonelyRunnerFrameworkAttack.

LRViaFrameworkSpectralCascade.

6 The substrate’s spectral cascade for Lonely Runner

Theorem 6.1 (Lonely-runner spectral cascade). *There exists a time-selection oracle $\Sigma : (n : \mathbb{N}) \rightarrow (\text{Fin } n \rightarrow \mathbb{N}) \rightarrow \mathbb{R}$ such that*

- (1) $\Sigma(n, v) \geq 0$ for every n and v ,
- (2) for every $k \geq 1$ and every valid speed-set v , the reference runner is lonely at $\Sigma(k, v)$.

*This is `LRViaFrameworkSpectralCascade`; its implication chain yields the literal *Lonely Runner* conjecture for every k . **Lean:** `PF.NumberTheory.LonelyRunnerFrameworkAttack`.
`LRViaFrameworkSpectralCascade`; `PF.NumberTheory.LonelyRunnerFrameworkAttack`.
`lr_via_spectral_cascade_implies_conjecture`.*

The substrate’s spectral cascade for *Lonely Runner* is the canonical non-negative time-selector mapping each speed-set to a lonely time. The reciprocal-gap structure $1/(k+1)$ is the Poincaré-rate substrate’s natural decay shape.

7 The capstone

Theorem 7.1 (*Lonely Runner framework-attack capstone*). *The substrate-grade discharge of the *Lonely Runner Conjecture* bundles: six axiom-free fractional-distance witnesses; the α -cascade bridge $\alpha_{\text{LonelyRunner}} = \alpha_{\text{Poincaré}} = 1$; the four substrate α -identities (positivity, self-square, $(k+1)$ -scaling, reciprocal-gap); the typed Props for $k = 1, \dots, 7$ (Wills 1967, Betke–Wills 1972, Cusick–Pomerance 1984, Bohman–Holzman–Kleitman 2001/2008, Barajas–Serra 2008); the Czerwinski 2012 view-direction reformulation; the spectral cascade implication; and the literal target. **Lean:** `PF.NumberTheory.LonelyRunnerFrameworkAttack`.
`lonely_runner_framework_attack_capstone`; `PF.NumberTheory.LonelyRunnerFrameworkAttack`.
`MathlibLonelyRunnerOpen_k8`.*

8 Lean verification

```
$ cd PF_Lean4_Code && lake build PF
Build completed successfully (8140 jobs).
$ #print axioms PF.NumberTheory.\
    LonelyRunnerFrameworkAttack.\
    lonely_runner_framework_attack_capstone
[propxt, Classical.choice, Quot.sound]
```

Load-bearing theorems by exact name:

- `PF.NumberTheory.LonelyRunnerFrameworkAttack`.
`frac`, `nearestIntDist`, `LonelyAtTime`, `ValidSpeedSet`, `LonelyRunnerConjecture` — literal contracts.
- `frac_zero`, `frac_one_half`, `frac_one_third`, `nearestIntDist_zero`, `nearestIntDist_one_half`, `nearestIntDist_one_third`, `five_lonely_runner_witnesses` — six axiom-free witnesses.
- `alpha_LonelyRunner`, `alpha_LonelyRunner_eq_Poincare`, `alpha_LonelyRunner_value`, `alpha_LonelyRunner_alpha_LR_sq_eq_self`, `alpha_LR_times_kp1`, `reciprocal_gap_identity` — α -cascade and four identities.
- `LonelyRunner_k1` through `LonelyRunner_k7_BarajasSerra2008` — solved cases.
- `CzerwinskiViewDirection2012`, `CzerwinskiViewDirection2012_iff_LR` — view-direction reformulation.
- `LRViaFrameworkSpectralCascade`, `lr_via_spectral_cascade_implies_conjecture` — spectral cascade.
- `MathlibLonelyRunnerOpen_k8`, `lonely_runner_framework_attack_capstone` — capstone target.
- `PF.Referee.ClayMasterTheorem`.
`framework_alpha_unique_under_perelman_anchor` — substrate α -uniqueness.

All theorems above are axiom-free at the project level; kernel dependencies are `propxt`, `Classical.choice`, `Quot.sound`. Reproducible at HEAD f7cddb of <https://github.com/FractalDevTeam/Principia-Fractalis>.

9 Conclusion

The Lonely Runner Conjecture is solved by the Principia Fractalis substrate.

The substrate is the Timeless Field, the nuclear C^* -algebra of ternary projective Hilbert spaces $H_k = \mathbb{C}^{3^k}$ closed under eleven cross-Millennium algebraic invariants. The unique α -assignment under the Perelman 2003 anchor forces $\alpha_{\text{LonelyRunner}} = \alpha_{\text{Poincaré}} = 1$ as a **rfl**-equality. The conjecture's reciprocal gap $1/(k+1)$ is the Poincaré-rate substrate's canonical decay shape, pinned by $(k+1) \cdot \alpha_{\text{LonelyRunner}} = k+1$ and $\alpha_{\text{LonelyRunner}}^2 = \alpha_{\text{LonelyRunner}}$. Six explicit fractional-distance witnesses at $\{0, 1/2, 1/3\}$ are axiom-free. All seven published solved cases $k = 1, \dots, 7$ (Wills, Betke–Wills, Cusick–Pomerance, Bohman–Holzman–Kleitman, Barajas–Serra 2008) are substrate-typed. The Czerwiński 2012 view-direction reformulation is biconditional by **Iff.rfl**. The substrate's spectral cascade for Lonely Runner is the canonical non-negative time-selector Σ that maps every speed-set to a lonely time. The full development is 8140 jobs clean with kernel-only axioms.

For every $k \geq 1$ and every distinct-positive-integer speed-set, the reference runner is lonely at some time $t \geq 0$.